Comment on the second-preimage resistance of the WChain FChain mode

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Abstract

WChain iterates a feed-forward compression function $\mathrm{CF}(V, B) = P_B(V) \oplus V$ in its FChain mode, with a two-step feedback, a final XOR checksum block, 10^* padding without a length field, and all-zero initial chaining values. We show that a four-block prefix $X \,\|\, Y \,\|\, Y \,\|\, X$ with $P_X(0) = P_Y^{-1}(0)$ returns both chaining registers to zero and adds nothing to the checksum. Hence $\mathrm{Hash}(X \,\|\, Y \,\|\, Y \,\|\, X \,\|\, M) = \mathrm{Hash}(M)$ for every message M, including one-block messages. A suitable pair (X, Y) is found by matching forward and inverse permutation calls, in about $2^{b/2}$ calls up to a small constant factor, and one pair serves every target. This gives second preimages for Version I in at most 2^{292} and for Version II in at most 2^{580} forward-call equivalents, against required levels of 2^{512} and 2^{1024}. The cost is also below the generic second-preimage estimate given in the specification. The attack is generic in the permutation, and it leaves the collision claim intact. We confirmed it on the full-size compression functions, on the full WChain round function at reduced lane width, and on ideal permutations.

Disclaimer. The second-preimage attack described in this note was produced by the author. It was subsequently verified with the assistance of AI, and all attack code (the reimplementation of both WChain versions and their inverse permutations, the comparison with the reference implementation, the reduced-width attack, and the ideal-permutation experiment) was written by AI. This note was itself drafted in its entirety by AI.

1 Summary

WChain [1] is a hash family submitted to the NGCC program. Version I has digest length $h = 512$, chaining state $b = 576$, message block $\mu = 1152$ and $R = 18$ rounds. Version II has $h = 1024$, $b = 1152$, $\mu = 2304$ and $R = 24$. The compression function is $\mathrm{CF}(V, B) = X_R \oplus \widehat{B}_R \oplus V$, where X_R is the state after R rounds started from $X_0 = V$ with the expanded blocks $\widehat{B}_0, \ldots, \widehat{B}_{R-1}$ injected, and $\widehat{B}_R$ is the last expanded block (eqs. (10) and (11) of [1]). We write

$$P_B(V) = X_R \oplus \widehat{B}_R, \qquad \text{so that} \qquad \mathrm{CF}(V, B) = P_B(V) \oplus V.$$

This P_B is the F_B of eq. (38). For each block B it is a permutation of $\{0,1\}^b$. Its inverse is cheap, because every round component is invertible and the round keys depend on B alone.

Our result is the following statement.

> If $P_X(0) = P_Y^{-1}(0)$, then $\mathrm{Hash}(X \,\|\, Y \,\|\, Y \,\|\, X \,\|\, M) = \mathrm{Hash}(M)$ for every message M.

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The table compares the cost of finding such a pair with the generic estimate of Section 5.1.4 of [1] and with the NGCC requirement of second-preimage resistance at the digest length. Costs are counted in forward calls of P_B, that is, in compression-function calls.

Version	h	b	this attack	estimate (50), every k	required
I	512	576	$\leq 2^{292}$	$\geq 2^{298.2}$	2^{512}
II	1024	1152	$\leq 2^{580}$	$\geq 2^{587.2}$	2^{1024}

One pair (X, Y) gives a second preimage of every message, with no long target, no checksum correction, and no property of the round function beyond an efficient inverse.

2 The FChain mode

The message is padded as $M^\star = M \,\|\, 1 \,\|\, 0^k$ with k minimal such that $|M| + 1 + k \equiv 0 \pmod{\mu}$, with no encoded length (eqs. (2) and (3)), and parsed as $M_1 \,\|\, \cdots \,\|\, M_l$. The initial values are $H_{-1} = 0$ and $H_0 = IV$ (eq. (6)), and the specification fixes all-zero initial values (Section 6.1); the reference code also starts both registers at zero. The hash is

$$H_{-1} = H_0 = 0, \qquad H_i = \mathrm{CF}(H_{i-1}, M_i) \oplus H_{i-2} \quad (1 \leq i \leq l),$$
$$\Sigma = M_1 \oplus M_2 \oplus \cdots \oplus M_l, \qquad Z = \mathrm{Trunc}_h\big(\mathrm{CF}(H_l, \Sigma)\big),$$

following eqs. (5), (7), (8) and (9). It helps to view one block as a map on the register pair. Block B maps $(H_{i-2}, H_{i-1}) = (u, v)$ to

$$T_B(u, v) = \big(v,\ \mathrm{CF}(v, B) \oplus u\big).$$

This is a Feistel round followed by a swap. With $\sigma(u, v) = (v, u)$ one checks that $\sigma \circ T_B \circ \sigma = T_B^{-1}$. So in a palindromic block sequence the second half undoes the first half, up to σ, and the attack uses exactly this.

3 The attack

Lemma 1. Let X, Y be blocks and (u, v) a register pair with

$$P_X(v) \oplus v \oplus u = P_Y^{-1}(v). \qquad (*)$$

Then the blocks X, Y, Y, X map (u, v) to (v, u).

Proof. Let $w = \mathrm{CF}(v, X) \oplus u = P_X(v) \oplus v \oplus u$. Condition $(*)$ says $P_Y(w) = v$, hence $\mathrm{CF}(w, Y) = w \oplus v$. The four steps are

$$(u, v) \xrightarrow{X} (v, w) \xrightarrow{Y} (w,\ \mathrm{CF}(w, Y) \oplus v) = (w, w)$$
$$\xrightarrow{Y} (w,\ \mathrm{CF}(w, Y) \oplus w) = (w, v) \xrightarrow{X} (v,\ \mathrm{CF}(v, X) \oplus w) = (v, u),$$

where the last step uses $\mathrm{CF}(v, X) = w \oplus u$. □

The zero start. The hash starts from $(H_{-1}, H_0) = (0, 0)$, which σ fixes. With $u = v = 0$, condition $(*)$ becomes

$$P_X(0) = P_Y^{-1}(0) =: a, \qquad \text{equivalently} \qquad \mathrm{CF}(0, X) = a \ \text{ and } \ \mathrm{CF}(a, Y) = a,$$

and the chaining values over the prefix are as follows.

Block	Resulting chaining value
X	$H_1 = \mathrm{CF}(0, X) \oplus H_{-1} = a$
Y	$H_2 = \mathrm{CF}(a, Y) \oplus H_0 = a$
Y	$H_3 = \mathrm{CF}(a, Y) \oplus H_1 = 0$
X	$H_4 = \mathrm{CF}(0, X) \oplus H_2 = 0$

Both registers return to zero. The value $a = P_Y^{-1}(0)$ is the classical Davies-Meyer fixed point of $\mathrm{CF}(\cdot, Y)$ [2, 3]. In plain Merkle-Damgård such a fixed point lets one repeat Y for free. The two-step feedback prevents that, since a second Y sends the register to 0, but the closing X then returns both registers to zero.

Theorem 1. If $P_X(0) = P_Y^{-1}(0)$, then for every message M the message

$$M' = X \,\|\, Y \,\|\, Y \,\|\, X \,\|\, M$$

satisfies $M' \neq M$ and $\mathrm{Hash}(M') = \mathrm{Hash}(M)$.

Proof. Since $|M'| = |M| + 4\mu$, the padding of M' uses the same k, so

$$\mathrm{Pad}(M') = X \,\|\, Y \,\|\, Y \,\|\, X \,\|\, \mathrm{Pad}(M).$$

The prefix adds $X \oplus Y \oplus Y \oplus X = 0$ to the checksum, so $\Sigma' = \Sigma$. After the prefix the registers are $(0, 0) = (H_{-1}, H_0)$, so the remaining blocks produce the same H_l, and the final call $\mathrm{CF}(H_l, \Sigma)$ is the same. The two messages differ in length. □

For a one-block target the second preimage has five padded blocks.

Remark (other initial values). One palindrome suffices because the zero start is fixed by σ, but the attack does not depend on it. From $(0, IV)$ with $IV \neq 0$, Lemma 1 with $P_X(IV) \oplus IV = P_Y^{-1}(IV)$ maps the pair to $(IV, 0)$, and a second palindrome Z, W, W, Z with $P_Z(0) \oplus IV = P_W^{-1}(0)$ maps it back to $(0, IV)$. The eight-block prefix again has zero checksum, and the cost doubles.

4 Cost

Take N_f blocks X and compute $P_X(0)$, and N_i blocks Y and compute $P_Y^{-1}(0)$. For an ideal family these are independent uniform b-bit values, so the expected number of matches is $N_f N_i / 2^b$. Each block has $\mu = 2b$ free bits, far more than needed. With $N_f = N_i = 2^{b/2}$ the work is about $2^{b/2+1}$ permutation calls and $2^{b/2}$ memory, and parallel collision search [4] removes the memory for a small constant factor.

An inverse call costs more than a forward call, mainly because the inverse linear layer is dense. In a direct word-level implementation it XORs 2569 rotated lanes per round in Version I and 5138 in Version II, against 45 and 90 in the forward Mix Words layer; we obtained these counts by inverting the binary matrix of the linear layer. Counting the word operations of the whole round, including the inverse S-box and the message expansion, one inverse call costs at most about $\rho = 60$ forward calls. Taking $N_f = 2^{b/2}\sqrt{\rho}$ and $N_i = 2^{b/2}/\sqrt{\rho}$ keeps one expected match, and the work becomes $2\sqrt{\rho}\, 2^{b/2} < 2^{b/2+4}$ forward calls. This is at most 2^{292} for Version I and 2^{580} for Version II.

5 Consequences

- *Universal.* The precomputation does not depend on the target. One pair (X, Y) gives a second preimage of every message, of any length, at no further cost, and $(X \,\|\, Y \,\|\, Y \,\|\, X)^j \,\|\, M$ collides with M for every $j \geq 1$.
- *Below the specification's own estimate.* The specification states no numeric second-preimage level. Its generic estimate for a 2^k-block target, eq. (50), is $C_{2\text{nd}}(k) \approx k2^{b/2+1} + 2^{b-k+1} + C_\Sigma$ with $C_\Sigma = (\mu + 1)2^{b/2}$ (eq. (43)). It is at least $C_\Sigma \approx 2^{298.2}$ for Version I and $2^{587.2}$ for Version II for every k, and close to 2^{b-k+1} for short targets. The palindrome avoids the C_Σ term because it cancels the checksum for free, and it avoids the 2^{b-k+1} term because it never connects into the target.
- *Not a collision attack.* The pair (M, M') is also a collision, but $2^{b/2} > 2^{h/2}$, so the claimed collision level $2^{h/2}$ stands.
- *The wrapper itself.* Sections 3.2 and 4.1 of [1] propose FChain as an upgrade path for SM3-like or SHA-2-like compression functions. Lemma 1 uses only the XOR feedback, the XOR checksum and the inverse of P_B. So around any compression function of the form $P_B(V) \oplus V$ with an efficiently invertible P_B, such as an SM3-like one, the wrapper gives at most about $2^{n/2}$ second-preimage resistance for an n-bit chaining value, with any initial value.

6 Verification

Reference agreement and full-size checks. We reimplemented both versions from the specification. Our compression function and hash agree with the submitted reference implementation on random inputs and on eleven message lengths around the block boundaries, for each version. We computed P_B^{-1} round by round and checked $P_B^{-1}(P_B(v)) = v$. On the full-size compression functions of both versions we then checked, for random blocks, the Davies-Meyer fixed point $\text{CF}(P_Y^{-1}(0), Y) = P_Y^{-1}(0)$ and Lemma 1 itself: for random X, Y, v and $u = P_X(v) \oplus v \oplus P_Y^{-1}(v)$, the blocks X, Y, Y, X map (u, v) to (v, u). Only the $2^{b/2}$ search for the zero start is out of reach at full size.

Full round function at reduced lane width. To run the complete attack, we kept the WChain round function and reduced only the lane width from 64 to w bits, so $b = 9w$ for Version I and $b = 18w$ for Version II. The S-box, the wiring π, the Mix Words support and rotation tables (offsets taken mod w), the Version II cross-swap, the message expansion (rotations mod w), the round counts $R = 18$ and $R = 24$, the mode with zero initial values, 10^* padding, two-step feedback and checksum block, and the truncation to the leftmost $8w$ or $16w$ bits are as specified. The round constants are the specified 64-bit lane constants folded to w bits by XOR. The linear layer is invertible for every width we tried. Each trial alternated forward and inverse calls until the first match $P_X(0) = P_Y^{-1}(0)$, and then hashed random messages M, of random bit length up to 5μ, with and without the prefix. The first match is expected after about $\sqrt{\pi}\, 2^{b/2}$ calls, that is $\log_2 \approx b/2 + 0.83$.

Version	w	b	trials	$\log_2$ mean calls	$b/2 + 0.83$	second preimages
I	2	18	300	9.84	9.83	60 000 / 60 000
I	3	27	150	14.36	14.33	30 000 / 30 000
I	4	36	40	19.02	18.83	4 000 / 4 000
I	5	45	16	23.19	23.33	1 600 / 1 600
II	1	18	300	9.88	9.83	60 000 / 60 000
II	2	36	40	18.70	18.83	4 000 / 4 000

At $w = 1$ all rotations vanish, so that row tests the round structure without them. The fitted slope of $\log_2$ mean calls against b is 0.497 for Version I, and 0.496 with Version II included, against the predicted 0.5. Every pair returned both registers to zero and gave $\mathrm{Hash}(X \,\|\, Y \,\|\, Y \,\|\, X \,\|\, M) = \mathrm{Hash}(M)$ for every message tested. As a control we also hashed $X \,\|\, Y \,\|\, X \,\|\, Y \,\|\, M$, which has the same zero checksum but is not a palindrome. It matched only at the chance rate: 3 of 60 000 at $b = 18$ for Version I, where the digest has 16 bits, and none at larger b. Each run at $b = 45$ took between 8 and 74 seconds on one core.

Ideal permutations. With lazily sampled independent random permutations P_B in the same mode, the search took $2^{6.82}$, $2^{8.84}$ and $2^{10.91}$ calls on average at $b = 12, 16, 20$ (300, 300 and 200 trials), against the predicted $2^{6.83}$, $2^{8.83}$ and $2^{10.83}$. Every pair gave second preimages for all messages tested.

7 Recommendation

1. Make the FChain iteration position-dependent or length-aware. A block counter in each compression call, as in HAIFA [5], or the message length in the final block, stops the prefix attack of this note. Changing the initial value alone does not, as the remark in Section 3 shows.
2. Revise the second-preimage discussion of Section 5.1.4 of [1]. With the current mode, second-preimage resistance is at most about $2^{b/2}$ for every target length, that is about 2^{288} and 2^{576} rather than 2^{512} and 2^{1024}.

Reproducibility. A reimplementation of both WChain versions (forward and inverse permutation, compression function and hash), the comparison with the submitted reference code, the reduced-width attack, and the ideal-permutation experiment are available on request.

Scope. This is a mode-level comment. The attack uses P_B only as a black box with an efficient inverse, so it holds for an ideal permutation family. It makes no claim about the WChain round function or the message expansion, and it is independent of the message-expansion properties reported in [6]. It does not affect the collision claim.

References

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